



Model Validation Report

Weight of Evidence logistic regression scorecard estimated and validated on resolved consumer loan outcomes. Feature ablation, discrimination, calibration, population stability, adverse action reason codes, stated limitations and reproduction instructions.

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Summary of results

Variant B, with the bureau credit score and all long horizon repayment history removed, reached 0.6883 area under the curve against 0.6874 for the platform internal risk grade on the same withheld vintages. Variant D, with every credit report field removed, reached 0.6535 on eight application level characteristics, retaining 81.9 percent of benchmark Gini.

Measure	Value	Basis
Loans used	1,249,246	Vintages 2012 to 2017, all resolved to repayment or charge off
Realized defaults	254,055	Observed outcomes, not modelled
Estimation set	786,820	Vintages 2012 to 2015, default rate 18.64 percent
Withheld set	462,426	Vintages 2016 to 2017, default rate 23.23 percent, out of time
Benchmark AUC	0.6874	Platform internal risk grade, same withheld set
Variant B AUC	0.6883	Score and long horizon history removed, 31 characteristics
Variant D AUC	0.6535	All credit report fields removed, 8 characteristics
Variant B KS	0.2707	Meets the working standard in the Technical Architecture document
Default rate, band 540 and below	44.61 percent	Withheld set
Default rate, band 620 to 640	2.12 percent	Withheld set, monotonic across all bands
Population stability index	0.0081	Estimation against withheld distribution

The measurement relevant to an applicant who cannot be scored conventionally is Variant D. Removing every credit report field and retaining only characteristics stated on an application preserves 81.9 percent of benchmark Gini, which bounds how much assessment capability survives without a credit history before any alternative data is introduced.



Three qualifications are material and are stated in full at section 7. The population is credit visible, so removing fields approximates an applicant without a file rather than replicating one. Requested loan term carries higher information value than any borrower characteristic and is a product selection. Calibration drifted between estimation and test eras while rank ordering held, which requires recalibration of the intercept on any deployment population.

1. Population and split design

Source	LendingClub accepted loan release, 2007 through 2018 quarter four		
Rows in release	2,260,701		
Matured in release	1,345,350 loans, 268,599 realized defaults		
Modelling window	Origination vintages 2012 through 2017		
Loans used	1,249,246, containing 254,055 realized defaults		
Excluded from modelling	96,104 loans in vintages outside the window		

TRAIN	vintages 2012-2015	786,820 loans	default rate 18.64 percent
TEST	vintages 2016-2017	462,426 loans	default rate 23.23 percent

The test set is out of time. Later vintages were withheld from estimation entirely and were not used for binning, coefficient fitting or feature selection. This is not a random split.

2. Leakage control

Excluded from every feature set: all post origination payment, recovery, settlement and hardship fields, and the refreshed bureau score. The originating platform internal risk grade, sub grade and interest rate were also excluded from every model and retained solely as an independent benchmark measured on the same withheld loans.

3. Feature ablation results

Five feature sets were fitted on identical training data and measured on identical withheld data. The ablation isolates how much of the models ranking power depends on the bureau credit score, on long horizon repayment history, on credit report data generally, and on requested loan term.

Variant	Feature set	k	AUC	Gini	vs bench (AUC)
Bench	Platform internal risk grade	-	0.6874	0.3748	-
A	Full file, includes score and deep history	51	0.6957	0.3914	+0.0083
B	Score and deep history removed	31	0.6883	0.3766	+0.0009
C	B with requested term also removed	30	0.6756	0.3512	-0.0118
D	All credit report fields removed	8	0.6535	0.3070	-0.0339



E	D with requested term also removed	7	0.6377	0.2754	-0.0497
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Variant B additional measures: Kolmogorov Smirnov 0.2707

Variant D feature list, application level only: reported income, debt to income ratio, employment length, housing status, requested amount, stated purpose, requested term, and extent of income verification.

4. What the ablation establishes

1. The incremental contribution of the bureau credit score and all long horizon repayment history is 0.0074 AUC, from 0.6957 to 0.6883. The score contributes less incremental separation than is commonly assumed.
2. Variant B matches the platform internal risk grade at +0.0009 AUC. That result is not evidence of thin file capability, because Variant B retains current credit report fields including revolving utilization, recent inquiries, bankcard limits and public records. A credit invisible applicant would have none of them. Section 7 limitation 7 records this.
3. Requested loan term carries information value 0.2448, against 0.0831 for the next characteristic. Removing term moves Variant B from +0.0009 to -0.0118 against the benchmark. A material share of Variant B ranking power derives from term, which is a product selection and not a borrower characteristic. This is recorded as a finding in the model card.
4. Variant D removes every credit report field and retains 81.9 percent of the benchmark Gini using eight application level characteristics. This is the result most relevant to an applicant with no credit file, and it is measured on a credit visible population, so it approximates that applicant without replicating one.
5. The Variant D result depends in part on requested term. Removing term from Variant D leaves Variant E at 0.6377 AUC and 0.2754 Gini, which is 73.5 percent of the benchmark Gini on seven application level characteristics. Where term is treated as a policy and pricing dimension rather than a borrower characteristic, Variant E is the figure to rely on.

5. Calibration, Variant B, out of time deciles

decile	n	predicted	observed
1	46243	4.78%	6.48%
2	46243	7.56%	10.94%
3	46242	9.66%	13.92%
4	46243	11.67%	17.28%
5	46242	13.76%	19.82%
6	46243	16.11%	23.03%
7	46242	18.94%	26.54%
8	46243	22.71%	30.45%
9	46242	28.71%	35.75%
10	46243	42.36%	48.09%

Deciles are strictly monotonic in realized default rate with no reversals. Predicted probability understates realized default in every decile. The training era default rate was 18.64 percent



against 23.23 percent in the test era. Rank ordering transfers out of time. The intercept requires recalibration on any deployment population before use in a live decision.

APPENDIX A - WEIGHT OF EVIDENCE BIN TABLES

Bins fitted on TRAIN vintages 2012-2015 only. Applied unchanged to test.

WoE = $\ln(P(\text{non-default in bin})/P(\text{default in bin}))$. Higher WoE = lower risk.

Points use PDO=20, base 600 at 20:1 odds. Higher points = lower risk.

term IV=0.2448 beta=-0.961

bin	n	dflt%	WoE	points
36 months	589488	14.03	0.3387	27.3
60 months	197332	32.39	-0.7377	-2.5

acc_open_past_24mths IV=0.0831 beta=-0.6963

bin	n	dflt%	WoE	points
(-1e+18, 1.0]	110577	12.67	0.4582	27.1
(1.0, 2.0]	106777	14.53	0.2998	24.0
(2.0, 3.0]	118624	16.5	0.1491	20.9
(3.0, 4.0]	111580	18.25	0.0274	18.5
(4.0, 5.0]	94509	19.78	-0.0721	16.5
(5.0, 6.0]	73587	21.37	-0.1694	14.5
(6.0, 8.0]	91262	23.56	-0.2953	12.0
(8.0, 1e+18]	72409	27.68	-0.5118	7.7

dti IV=0.0735 beta=-0.6176

bin	n	dflt%	WoE	points
(-1e+18, 8.45]	98429	13.2	0.4099	25.2
(8.45, 11.99]	98337	14.56	0.2959	23.2
(11.99, 14.9]	98385	15.74	0.2039	21.6
(14.9, 17.72]	98511	17.36	0.0866	19.5
(17.72, 20.66]	98427	18.67	-0.0025	17.9
(20.66, 24.05]	98243	20.51	-0.1193	15.8
(24.05, 28.46]	98231	22.48	-0.2358	13.7
(28.46, 1e+18]	98255	26.59	-0.4585	9.8

num_tl_op_past_12m IV=0.0585 beta=-0.3026

bin	n	dflt%	WoE	points
(-1e+18, 1.0]	326982	15.14	0.254	20.2
(1.0, 2.0]	175786	18.71	-0.0003	17.9
(2.0, 3.0]	125206	21.36	-0.1657	16.5
(3.0, 4.0]	71112	23.66	-0.2978	15.3
(4.0, 1e+18]	59993	26.69	-0.459	13.9

bc_open_to_buy IV=0.0546 beta=-0.3533

bin	n	dflt%	WoE	points
(-1e+18, 419.0]	96615	23.26	-0.2798	15.1
(419.0, 1202.0]	96418	21.43	-0.1743	16.2
(1202.0, 2334.0]	96450	20.89	-0.1418	16.5
(2334.0, 3941.0]	96467	19.92	-0.082	17.1
(3941.0, 6388.0]	96476	18.92	-0.0179	17.8
(6388.0, 10643.0]	96481	17.38	0.0853	18.8
(10643.0, 19966.0]	96466	15.47	0.2251	20.2
(19966.0, 1e+18]	96475	11.87	0.5313	23.4

verification_status IV=0.0521 beta=-0.2625



bin	n	dflt%	WoE	points
Not Verified	227170	13.49	0.3841	20.8
Source Verified	298513	20.01	-0.0883	17.3
Verified	261137	21.54	-0.181	16.6

avg_cur_bal IV=0.0431 beta=-0.2349

bin	n	dflt%	WoE	points
(-1e+18, 1884.0]	94891	20.46	-0.1115	17.2
(1884.0, 3070.0]	94898	21.5	-0.1744	16.8
(3070.0, 4564.0]	94866	21.32	-0.1637	16.8
(4564.0, 7238.0]	94884	20.95	-0.1414	17.0
(7238.0, 12062.25]	94878	19.67	-0.062	17.5
(12062.25, 18379.0]	94889	17.7	0.0672	18.4
(18379.0, 28323.75]	94877	15.57	0.2211	19.4
(28323.75, 1e+18]	94884	12.46	0.4803	21.2

mo_sin_rcnt_tl IV=0.041 beta=-0.2635

bin	n	dflt%	WoE	points
(-1e+18, 2.0]	160972	22.48	-0.2316	16.2
(2.0, 3.0]	79967	21.16	-0.1541	16.8
(3.0, 4.0]	70626	19.98	-0.0819	17.3
(4.0, 6.0]	114983	19.07	-0.0237	17.8
(6.0, 7.0]	48383	18.31	0.026	18.1
(7.0, 10.0]	104652	17.25	0.0987	18.7
(10.0, 15.0]	91859	15.98	0.1907	19.4
(15.0, 1e+18]	87637	12.83	0.4467	21.3

tot_hi_cred_lim IV=0.0386 beta=-0.5472

bin	n	dflt%	WoE	points
(-1e+18, 30264.0]	94889	20.51	-0.1148	16.1
(30264.0, 47700.0]	94953	21.15	-0.1532	15.5
(47700.0, 69529.0]	94818	21.45	-0.1711	15.2
(69529.0, 107743.0]	94881	21.07	-0.1487	15.6
(107743.0, 172398.0]	94885	19.5	-0.0512	17.1
(172398.0, 244780.0]	94884	17.46	0.0838	19.3
(244780.0, 350734.0]	94884	15.4	0.2342	21.6
(350734.0, 1e+18]	94885	13.1	0.4228	24.6

total_bc_limit IV=0.0382 beta=-0.5262

bin	n	dflt%	WoE	points
(-1e+18, 4500.0]	102201	22.11	-0.2128	14.7
(4500.0, 7500.0]	97239	20.97	-0.1456	15.7
(7500.0, 10600.0]	94200	20.9	-0.1413	15.8
(10600.0, 14400.0]	96165	19.95	-0.0826	16.7
(14400.0, 19500.0]	98604	18.76	-0.0064	17.8
(19500.0, 27000.0]	96803	17.6	0.0716	19.0
(27000.0, 40800.0]	97133	16.05	0.1823	20.7
(40800.0, 1e+18]	96980	12.83	0.4434	24.7

loan_amnt IV=0.0352 beta=-0.7273

bin	n	dflt%	WoE	points
(-1e+18, 5500.0]	100465	14.6	0.2923	24.1
(5500.0, 8000.0]	106207	14.72	0.283	23.9
(8000.0, 10000.0]	101549	16.56	0.143	20.9
(10000.0, 12500.0]	86611	19.89	-0.0809	16.2
(12500.0, 15950.0]	97032	20.09	-0.0933	16.0
(15950.0, 20000.0]	123168	20.94	-0.1452	14.9
(20000.0, 25000.0]	82383	21.32	-0.1683	14.4
(25000.0, 1e+18]	89405	21.73	-0.1921	13.9

tot_cur_bal IV=0.0335 beta=0.2366



bin	n	dflt%	WoE	points
(-1e+18, 16618.0]	94886	18.77	-0.0041	18.0
(16618.0, 29067.0]	94889	20.66	-0.1237	18.8
(29067.0, 45269.0]	94884	21.62	-0.1812	19.2
(45269.0, 77786.0]	94881	21.82	-0.1932	19.3
(77786.0, 139746.75]	94884	19.71	-0.0645	18.4
(139746.75, 205660.0]	94887	17.75	0.0639	17.5
(205660.0, 302126.25]	94883	15.76	0.2069	16.5
(302126.25, 1e+18]	94885	13.56	0.3834	15.3

annual_inc IV=0.0315 beta=-0.6973

bin	n	dflt%	WoE	points
(-1e+18, 36000.0]	103609	22.0	-0.2084	13.7
(36000.0, 45000.0]	96850	21.24	-0.1635	14.7
(45000.0, 55000.0]	112744	20.36	-0.1098	15.7
(55000.0, 64000.0]	80371	19.69	-0.0681	16.6
(64000.0, 75000.0]	109556	18.67	-0.0025	17.9
(75000.0, 90000.0]	96179	17.15	0.1011	20.0
(90000.0, 115000.0]	90186	15.43	0.2277	22.5
(115000.0, 1e+18]	97325	13.99	0.3424	24.8

mo_sin_rcnt_rev_tl_op IV=0.0305 beta=0.3046

bin	n	dflt%	WoE	points
(-1e+18, 2.0]	122769	22.33	-0.2227	19.9
(2.0, 4.0]	110207	20.46	-0.1114	18.9
(4.0, 6.0]	89138	19.55	-0.0547	18.4
(6.0, 8.0]	73538	19.24	-0.0347	18.2
(8.0, 11.0]	85054	18.69	0.0009	17.9
(11.0, 16.0]	96853	18.24	0.0308	17.7
(16.0, 26.0]	89334	16.12	0.1802	16.4
(26.0, 1e+18]	92185	13.55	0.3842	14.6

percent_bc_gt_75 IV=0.0303 beta=-0.5147

bin	n	dflt%	WoE	points
(-1e+18, 20.0]	209050	15.15	0.2494	21.6
(20.0, 33.3]	101809	17.14	0.1024	19.5
(33.3, 50.0]	129556	18.63	0.0009	18.0
(50.0, 66.7]	92355	20.22	-0.1008	16.4
(66.7, 80.0]	62020	21.6	-0.1842	15.2
(80.0, 1e+18]	176676	21.79	-0.1953	15.0

mths_since_recent_bc IV=0.0281 beta=-0.6162

bin	n	dflt%	WoE	points
(-1e+18, 3.0]	102659	22.25	-0.2221	14.0
(3.0, 6.0]	99329	20.49	-0.1172	15.9
(6.0, 10.0]	108679	19.82	-0.0755	16.6
(10.0, 14.0]	90011	19.87	-0.0787	16.5
(14.0, 20.0]	92615	18.67	-0.002	17.9
(20.0, 30.0]	91595	17.47	0.0796	19.4
(30.0, 53.0]	91546	16.08	0.1788	21.1
(53.0, 1e+18]	95954	13.94	0.3467	24.1

bc_util IV=0.0271 beta=-0.0033

bin	n	dflt%	WoE	points
(-1e+18, 27.7]	96724	14.84	0.2735	18.0
(27.7, 43.9]	96616	15.97	0.1868	18.0
(43.9, 56.6]	96037	17.42	0.0824	17.9
(56.6, 67.8]	96793	18.31	0.0222	17.9
(67.8, 78.0]	97239	19.03	-0.0251	17.9
(78.0, 87.1]	95532	19.58	-0.0606	17.9
(87.1, 94.8]	96732	20.76	-0.1337	17.9



(94.8, 1e+18]	95691	23.27	-0.2806	17.9
total_rev_hi_lim IV=0.0256 beta=-0.223				
bin	n	dflt%	WoE	points
(-1e+18, 9200.0]	95086	20.76	-0.1301	17.1
(9200.0, 13700.0]	95701	20.41	-0.1081	17.2
(13700.0, 18150.0]	93881	20.23	-0.0973	17.3
(18150.0, 23300.0]	95737	20.0	-0.0828	17.4
(23300.0, 29800.0]	94618	19.47	-0.0493	17.6
(29800.0, 39125.5]	94286	18.59	0.0076	18.0
(39125.5, 55800.0]	94981	16.79	0.1312	18.8
(55800.0, 1e+18]	94789	13.38	0.3985	20.5
mort_acc IV=0.0239 beta=-0.4718				
bin	n	dflt%	WoE	points
(-1e+18, 1.0]	442663	20.57	-0.121	16.3
(1.0, 2.0]	108683	17.53	0.076	19.0
(2.0, 3.0]	83039	16.3	0.1635	20.2
(3.0, 4.0]	59743	15.64	0.2133	20.8
(4.0, 1e+18]	85197	14.61	0.2932	21.9
mths_since_recent_inq IV=0.0228 beta=-0.5787				
bin	n	dflt%	WoE	points
(-1e+18, 1.0]	143459	22.72	-0.2076	14.5
(1.0, 2.0]	62999	21.04	-0.1095	16.1
(2.0, 3.0]	56971	20.62	-0.084	16.5
(3.0, 5.0]	96909	19.46	-0.0116	17.7
(5.0, 7.0]	79992	18.54	0.0482	18.7
(7.0, 10.0]	90645	17.86	0.0942	19.5
(10.0, 15.0]	94775	16.9	0.1607	20.6
(15.0, 1e+18]	74266	15.45	0.2679	22.4
home_ownership IV=0.0223 beta=-0.7144				
bin	n	dflt%	WoE	points
MORTGAGE	391753	16.44	0.1523	21.1
OWN	78358	19.17	-0.0346	17.2
RENT	316618	21.23	-0.1625	14.6
purpose IV=0.0196 beta=-0.6141				
bin	n	dflt%	WoE	points
car	6922	14.09	0.3338	23.9
credit_card	185814	15.59	0.215	21.7
debt_consolidation	468555	19.85	-0.0781	16.6
home_improvement	45310	16.78	0.1273	20.2
house	3148	21.44	-0.176	14.8
major_purchase	14076	17.19	0.0985	19.7
medical	7469	19.73	-0.0712	16.7
moving	4640	21.83	-0.1988	14.4
other	36824	19.84	-0.0775	16.6
small_business	8002	27.14	-0.4867	9.3
vacation	4261	17.46	0.0789	19.3
wedding	1343	13.55	0.3768	24.6
revol_util IV=0.0189 beta=-0.385				
bin	n	dflt%	WoE	points
(-1e+18, 25.9]	98693	14.65	0.2887	21.1
(25.9, 37.8]	98817	16.6	0.1406	19.5
(37.8, 47.2]	97654	17.85	0.0525	18.5
(47.2, 55.8]	98657	18.85	-0.014	17.8
(55.8, 64.2]	97908	19.36	-0.0471	17.4
(64.2, 73.2]	98457	20.04	-0.0901	16.9



(73.2, 83.9]	98611	20.62	-0.1258	16.5
(83.9, 1e+18]	97629	21.15	-0.1582	16.2
inq_last_6mths IV=0.0187 beta=-0.4208				
bin	n	dflt%	WoE	points
(-1e+18, 1.0]	656469	17.7	0.0632	18.7
(1.0, 2.0]	83834	22.56	-0.2405	15.0
(2.0, 1e+18]	46517	24.81	-0.3652	13.5
open_acc IV=0.0076 beta=0.1534				
bin	n	dflt%	WoE	points
(-1e+18, 6.0]	112720	16.47	0.1497	17.3
(6.0, 8.0]	127837	17.47	0.0786	17.6
(8.0, 9.0]	71504	18.22	0.0277	17.8
(9.0, 11.0]	134646	18.7	-0.0045	18.0
(11.0, 12.0]	58082	19.24	-0.0393	18.1
(12.0, 14.0]	93422	19.37	-0.0475	18.2
(14.0, 18.0]	109599	19.72	-0.0701	18.3
(18.0, 1e+18]	79010	21.05	-0.1519	18.6
emp_length IV=0.0067 beta=-0.9951				
bin	n	dflt%	WoE	points
1 year	50242	18.81	-0.0112	17.6
10+ years	259619	17.8	0.056	19.5
2 years	69416	18.2	0.0288	18.8
3 years	61396	18.42	0.014	18.3
4 years	45744	18.49	0.0094	18.2
5 years	49010	18.3	0.0223	18.6
6 years	38171	18.53	0.0072	18.1
7 years	40079	18.61	0.0018	18.0
8 years	39462	19.14	-0.0329	17.0
9 years	30824	19.25	-0.0399	16.8
< 1 year	61358	19.12	-0.0318	17.0
MISS	41499	23.71	-0.3053	9.2
revol_bal IV=0.0036 beta=1.0487				
bin	n	dflt%	WoE	points
(-1e+18, 4026.0]	98388	18.73	-0.0063	18.1
(4026.0, 6427.0]	98331	18.77	-0.009	18.2
(6427.0, 8893.0]	98362	19.13	-0.0325	18.9
(8893.0, 11714.0]	98354	19.24	-0.0392	19.1
(11714.0, 15294.0]	98334	19.27	-0.0414	19.2
(15294.0, 20472.0]	98352	19.12	-0.0317	18.9
(20472.0, 29808.0]	98354	18.41	0.0148	17.5
(29808.0, 1e+18]	98345	16.41	0.1545	13.3
total_bal_ex_mort IV=0.0029 beta=0.7152				
bin	n	dflt%	WoE	points
(-1e+18, 12746.0]	97418	16.84	0.1249	15.4
(12746.0, 20683.0]	97420	18.36	0.0195	17.5
(20683.0, 28288.0]	97410	18.64	0.001	17.9
(28288.0, 36535.0]	97420	19.21	-0.0356	18.7
(36535.0, 46498.5]	97410	19.33	-0.0438	18.8
(46498.5, 60650.0]	97419	19.36	-0.0454	18.9
(60650.0, 86651.5]	97412	19.32	-0.0429	18.8
(86651.5, 1e+18]	97416	18.21	0.0297	17.3
tot_coll_amt IV=0.0015 beta=-1.002				
bin	n	dflt%	WoE	points
(-1e+18, 80.0]	664631	18.48	0.0149	18.4
(80.0, 1e+18]	94448	20.29	-0.1009	15.0



total_il_high_credit_limit	IV=0.0015	beta=1.437			
bin	n	dflt%	WoE	points	
(-1e+18, 13582.0]	189772	17.78	0.0623	15.4	
(13582.0, 21845.0]	94888	18.93	-0.0144	18.5	
(21845.0, 30067.0]	94887	18.8	-0.0063	18.2	
(30067.0, 40055.0]	94882	19.11	-0.0266	19.0	
(40055.0, 53971.0]	94881	19.4	-0.0453	19.8	
(53971.0, 78278.0]	94885	19.19	-0.0315	19.2	
(78278.0, 1e+18]	94884	18.66	0.003	17.8	

pub_rec	IV=0.0009	beta=-0.4527			
bin	n	dflt%	WoE	points	
(-1e+18, 1.0]	765970	18.56	0.0051	18.0	
(1.0, 1e+18]	20850	21.45	-0.1758	15.6	

APPENDIX B - SCORECARD SUMMARY AND COEFFICIENTS

Model: logistic regression on Weight of Evidence transformed characteristics

Characteristics: 31

Intercept: -1.476631

Scaling: PDO=20, base score 600 at 20:1 good:bad odds, factor=28.8539, offset=513.5614

characteristic	IV	beta	pts_min	pts_max	range
term	0.2448	-0.961	-2.5	27.3	29.8
acc_open_past_24mths	0.0831	-0.6963	7.7	27.1	19.5
dti	0.0735	-0.6176	9.8	25.2	15.5
num_tl_op_past_12m	0.0585	-0.3026	13.9	20.2	6.2
bc_open_to_buy	0.0546	-0.3533	15.1	23.4	8.3
verification_status	0.0521	-0.2625	16.6	20.8	4.3
avg_cur_bal	0.0431	-0.2349	16.8	21.2	4.4
mo_sin_rcnt_tl	0.041	-0.2635	16.2	21.3	5.2
tot_hi_cred_lim	0.0386	-0.5472	15.2	24.6	9.4
total_bc_limit	0.0382	-0.5262	14.7	24.7	10.0
loan_amnt	0.0352	-0.7273	13.9	24.1	10.2
tot_cur_bal	0.0335	0.2366	15.3	19.3	3.9
annual_inc	0.0315	-0.6973	13.7	24.8	11.1
mo_sin_rcnt_rev_tl_op	0.0305	0.3046	14.6	19.9	5.3
percent_bc_gt_75	0.0303	-0.5147	15.0	21.6	6.6
mths_since_recent_bc	0.0281	-0.6162	14.0	24.1	10.1
bc_util	0.0271	-0.0033	17.9	18.0	0.1
total_rev_hi_lim	0.0256	-0.223	17.1	20.5	3.4
mort_acc	0.0239	-0.4718	16.3	21.9	5.6
mths_since_recent_inq	0.0228	-0.5787	14.5	22.4	7.9
home_ownership	0.0223	-0.7144	14.6	21.1	6.5
purpose	0.0196	-0.6141	9.3	24.6	15.3
revol_util	0.0189	-0.385	16.2	21.1	5.0
inq_last_6mths	0.0187	-0.4208	13.5	18.7	5.2
open_acc	0.0076	0.1534	17.3	18.6	1.3
emp_length	0.0067	-0.9951	9.2	19.5	10.4
revol_bal	0.0036	1.0487	13.3	19.2	5.9
total_bal_ex_mort	0.0029	0.7152	15.4	18.9	3.5
tot_coll_amt	0.0015	-1.002	15.0	18.4	3.3
total_il_high_credit_limit	0.0015	1.437	15.4	19.8	4.5
pub_rec	0.0009	-0.4527	15.6	18.0	2.4



SCORE DISTRIBUTION

train min=477.0 p5=521.4 median=562.4 p95=597.2 max=647.6

test min=477.7 p5=524.6 median=563.9 p95=598.8 max=651.3

DEFAULT RATE BY SCORE BAND (test, out of time)

(0, 540]	n= 68782	default= 44.61%
(540, 560]	n= 128682	default= 29.13%
(560, 580]	n= 162276	default= 18.4%
(580, 600]	n= 82206	default= 10.26%
(600, 620]	n= 18955	default= 4.96%
(620, 640]	n= 1512	default= 2.12%
(640, 9999]	n= 13	default= 0.0%

APPENDIX C - POPULATION STABILITY INDEX

PSI = sum over bands of (actual pct - expected pct) * ln(actual pct / expected pct)

Expected = TRAIN score distribution (2012-2015). Bands are train score deciles.

Interpretation: below 0.10 stable, 0.10 to 0.25 moderate shift, above 0.25 significant.

score band	expected%	actual%	contrib
(-1e+18, 530.446]	10.0	7.9	0.00493
(530.446, 542.208]	10.0	9.06	0.00093
(542.208, 550.357]	10.0	9.89	1e-05
(550.357, 556.783]	10.0	10.13	2e-05
(556.783, 562.421]	10.0	10.23	5e-05
(562.421, 567.85]	10.0	10.17	3e-05
(567.85, 573.559]	10.0	10.29	8e-05
(573.559, 580.211]	10.0	10.43	0.00018
(580.211, 589.528]	10.0	10.67	0.00043
(589.528, 1e+18]	10.0	11.24	0.00145

TOTAL PSI train vs test (2016-2017): 0.0081

PSI BY VINTAGE (expected = train)

2012 n= 53367 PSI=0.0322

2013 n= 134804 PSI=0.0053

2014 n= 223103 PSI=0.0015

2015 n= 375546 PSI=0.0008

2016 n= 293105 PSI=0.0061

2017 n= 169321 PSI=0.0131

APPENDIX D - ADVERSE ACTION REASON CODES ON REAL RECORDS

Regulation B section 1002.9 requires a statement of the specific principal reasons for adverse action. Reasons below are the characteristics on which the applicant lost the most scorecard points relative to the highest scoring bin for that



characteristic. Records are drawn from the held-out 2016-2017 test vintages.

APPLICANT 1 total score 477.7 modelled default probability 77.6%
realized outcome in data: DEFAULT
principal reasons:
1. Length of loan term requested [bin 60 months, 29.8 points below best]
2. Number of accounts opened in the last 24 months [bin (8.0, 1e+18], 19.5 points below best]
3. Ratio of debt obligations to income [bin (28.46, 1e+18], 15.5 points below best]
4. Length of employment reported [bin MISS, 10.4 points below best]

APPLICANT 2 total score 478.3 modelled default probability 77.22%
realized outcome in data: DEFAULT
principal reasons:
1. Length of loan term requested [bin 60 months, 29.8 points below best]
2. Number of accounts opened in the last 24 months [bin (8.0, 1e+18], 19.5 points below best]
3. Ratio of debt obligations to income [bin (28.46, 1e+18], 15.5 points below best]
4. Income reported on application [bin (-1e+18, 36000.0], 11.1 points below best]

APPLICANT 3 total score 478.5 modelled default probability 77.1%
realized outcome in data: DEFAULT
principal reasons:
1. Length of loan term requested [bin 60 months, 29.8 points below best]
2. Number of accounts opened in the last 24 months [bin (8.0, 1e+18], 19.5 points below best]
3. Ratio of debt obligations to income [bin (28.46, 1e+18], 15.5 points below best]
4. Income reported on application [bin (-1e+18, 36000.0], 11.1 points below best]

APPLICANT 4 total score 479.4 modelled default probability 76.55%
realized outcome in data: DEFAULT
principal reasons:
1. Length of loan term requested [bin 60 months, 29.8 points below best]
2. Number of accounts opened in the last 24 months [bin (8.0, 1e+18], 19.5 points below best]
3. Ratio of debt obligations to income [bin (28.46, 1e+18], 15.5 points below best]
4. Income reported on application [bin (-1e+18, 36000.0], 11.1 points below best]

APPLICANT 5 total score 480.3 modelled default probability 76.01%
realized outcome in data: DEFAULT
principal reasons:
1. Length of loan term requested [bin 60 months, 29.8 points below best]
2. Number of accounts opened in the last 24 months [bin (8.0, 1e+18], 19.5 points below best]
3. Ratio of debt obligations to income [bin (28.46, 1e+18], 15.5 points below best]
4. Length of employment reported [bin MISS, 10.4 points below best]

REASON CODE FREQUENCY across the lowest scoring 5000 test records

Length of loan term requested	4992	(99.8%)
Number of accounts opened in the last 24 months	4608	(92.2%)
Ratio of debt obligations to income	3950	(79.0%)
Income reported on application	1920	(38.4%)
Amount of credit requested	1512	(30.2%)
Time since most recent bankcard opened	903	(18.1%)
Total credit limit extended across accounts	662	(13.2%)
Length of employment reported	656	(13.1%)
Total bankcard credit limit	650	(13.0%)
Stated purpose of the loan	143	(2.9%)
Available credit on revolving bankcard accounts	4	(0.1%)

7. Stated limitations



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1. No cash flow, rental, utility or telecom variable exists in this source. Validation of those features is deferred to the retrospective phase.
 2. This source contains no race, ethnicity or sex field, so disparate impact could not be tested at this phase. Fair lending testing across protected classes is assigned to the retrospective phase against a partner institution borrower record set.
 3. Requested loan term carries the highest information value in every variant that retains it, and term is a product selection rather than a borrower characteristic. Results for variants retaining term should be read with that dependency in view. Variants C and E report the effect of removing it.
 4. Vintages from 2018 were excluded because term length exceeds the observation window, so resolved 2018 loans are biased toward early payoff and early default. Test vintages are not fully matured for 60 month terms.
 5. Calibration drifted. Training era default rate was 18.64 percent against 23.23 percent in the test era. Rank ordering transfers out of time. The intercept requires recalibration on any deployment population before use in a live decision.
 6. These applicants are not a CDFI applicant population. Coefficients must be refit on a partner institution portfolio before any live lending decision.
 7. This population is credit visible. Variants B and D reduce the feature set to approximate an applicant with limited or no credit file. They do not replicate one. Variant B in particular retains current credit report fields, so its parity with the benchmark is not evidence of capability on a credit invisible applicant.

8. Reproduction

<code>extract_matured.py</code>	builds the matured loan set from the public source release
<code>fit_scorecard.py</code>	fits Weight of Evidence bins and the model variants
<code>build_appendices.py</code>	produces Appendices A through D
<code>ablation.py</code>	fits and compares the five feature sets in section 3

Source: LendingClub accepted loan release, 2007 through 2018 quarter four, 2,260,701 rows. Bins are fitted on training vintages only and applied unchanged to withheld vintages.

Addendum to the Model Validation Report

Model Validation Report: Variant F, Debt to Income Ratio Removed

Get Credit Rate Foundation. Addendum dated 18 September 2026. Version 1.1 (supersedes Version 1.0 issued earlier the same day, which lettered this variant E in error). This addendum supplements the Model Validation Report; it does not replace any part of it. All original results, including Variant D, are unchanged.

1. Purpose

Variant D in the Model Validation Report removes the credit score and every credit history field and retains eight characteristics reported on the loan application: reported income, debt to income ratio, employment length, housing status, requested amount, stated purpose, requested term, and extent of income verification. In the public LendingClub release, the debt to income ratio is computed from monthly debt obligations reported to the credit bureaus divided by self-reported income. Because that ratio draws in part on credit bureau data, the author estimated a further variant, Variant F, with the debt to income ratio also removed. It is lettered F because the Model Validation Report already uses Variant E for the model with requested term removed, so that the reported result rests on characteristics that contain no credit bureau input.

2. Method

Identical to the Model Validation Report. Population: 1,249,246 resolved consumer loans from the public LendingClub release, origination vintages 2012 through 2017, containing 254,055 realized defaults. Out-of-time split: vintages 2012 to 2015 for estimation (786,820 loans), vintages 2016 to 2017 for testing (462,426 loans). Weight of Evidence binning on the estimation set (eight quantile bins for numeric characteristics, minimum bin size 1,000), logistic regression on the transformed characteristics, discrimination measured by area under the receiver operating characteristic curve (AUC) and Gini coefficient ($2 \times \text{AUC} - 1$) on the test set. Benchmark: the platform internal risk grade on the same test set. Variant D was re-estimated under the same code path to confirm reproduction before Variant F was estimated.

3. Results

Variant	Feature set	Characteristics	Out-of-time AUC	Gini	Share of benchmark Gini
Benchmark	Platform internal risk grade	n/a	0.6874	0.3749	100.0 percent
D (reproduced)	Application level only, as listed in Section 1	8	0.6535	0.3070	81.9 percent
F (new)	Variant D with debt to income ratio removed	7	0.6437	0.2874	76.7 percent

D minus F	Contribution of debt to income ratio	1	0.0098 AUC	0.0196 Gini	5.2 points
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Information value of the debt to income ratio on the estimation set: 0.0735, consistent with Appendix A of the Model Validation Report.

4. What the addendum establishes

With the credit score, every credit history field, and the debt to income ratio all removed, seven characteristics that contain no credit bureau input retain 76.7 percent of the ranking power of the platform internal risk grade on 462,426 out-of-time loans. The debt to income ratio contributes 5.2 percentage points of benchmark Gini. Variant D remains the reported result for an applicant who can state their existing obligations; Variant F is the reported result for an applicant about whom nothing from a credit bureau is available.

5. Limitation

Every borrower in the LendingClub population held a credit file. Both variants show that the scorecard architecture ranks repayment risk without bureau inputs on a credit-visible population; neither measures performance on borrowers who have no file. That measurement is the object of Phase 2 of the Foundation implementation plan.

6. Reproducibility

Script: `ablation_variant_f.py`, retained with the project source alongside the scripts listed at page 12 of the Model Validation Report. Input: `matured.parquet`, the same resolved loan set used in the Model Validation Report. Results file: `variantF_results.json`. Run date: 18 September 2026.